

Yoneda's Lemma and the Yoneda Perspective

Claudio Bastiani-Fonck

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Abstract

In my view, the Yoneda lemma is almost a “proof” that category theory is sufficiently strong to study diverse mathematical objects. It says that the morphisms encode all the information about the object that you are trying to study. This makes it the most important theorem in category theory (in my opinion).

Contents

1	Introduction	1
2	The Correspondence	3
3	Naturality	4
4	The Yoneda Embedding	4
5	Rambling	5

Introduction

If you've done a good amount of abstract algebra, you've probably done plenty of proofs using universal properties.

$\text{Hom}(M \otimes_R N, P) \cong \text{Bil}_R(M, N; P)$ 	see diagram
$\text{Hom}(\text{coker}(f), P) \cong \ker(g \mapsto f \circ g)$ 	$\text{Hom}(\varinjlim X_i, Y) \cong \varprojlim \text{Hom}(X_i, Y)$

Table 1: Popular universal properties

Here we can see some universal properties for commonly used mathematical constructions. One is encouraged to see which of these they recognize. Generally one “proves” a universal property by first constructing an object that satisfies it, then proving uniqueness. However, using category theory one may skip the uniqueness verification by citing Yoneda’s lemma. This note is a proof of that statement.

First, we see that all universal properties have the structure of specifying all the maps into or out of an object in terms of the other maps in the category. For example, a map out of $\text{coker}(f)$ is equivalently a map $Y \rightarrow P$ which is 0 when precomposed with f . Therefore we want to prove that specifying maps into or out of an object is enough to specify the object (up to isomorphism). To do this we introduce the hom-functor.

Definition 1.1. Given a locally small category $\mathcal{C}$ and $A \in \mathcal{C}$ we denote the *hom-functor* $h_A: \mathcal{C} \rightarrow \mathbf{Set}$ as the functor which sends:

- (a) $X \in \mathcal{C}$ to $\mathcal{C}(A, X)$, the set of morphisms from A to X ;
- (b) $f \in \mathcal{C}(X, Y)$ to $f \circ -: \mathcal{C}(A, X) \rightarrow \mathcal{C}(A, Y)$, which acts via post-composition by f on morphisms in $\mathcal{C}(A, X)$.

The hom-functor h_A encodes all the information about how A maps into the other objects of $\mathcal{C}$. From a philosophical perspective, we would like to say that understanding h_A is equivalent to understanding A . This legitimizes the categorical perspective that studying the morphisms between mathematical objects is the same as studying those objects.

From a mathematical perspective we claim that $X, Y \in \mathcal{C}$ are isomorphic iff their hom-functors h_X and h_Y are isomorphic ([Theorem 4.2](#)). Since universal properties completely specify the hom-functor, this lets us skip all the tedious proofs of uniqueness of universal properties.

The astute reader may point out that this only works on universal properties which specify the maps out of an object, and many universal properties instead specify the maps into an object. To this reader I leave as an exercise to recreate this entire note using instead the functor $h^A = \mathcal{C}(-, A)$.

So how should we go about proving this statement? Suppose $A \in \mathcal{C}$. We need some way of understanding the “structure” of A using the purely functorial information given to us by h_A . However, A is a purely abstract object which has no concrete structure. On the other hand, we can imbue A with some tangible structure by “representing” our category $\mathcal{C}$ with a more concrete set-theoretic mathematical object, similar

to how an abstract one-object category of a group can be represented by permutations on a set that is being acted on. So how do we formalize this?

Suppose that $F: \mathcal{C} \rightarrow \mathbf{Set}$ is some arbitrary set-valued functor. Set-valued functors can be thought of as “embeddings” or “representations” of our abstract category $\mathcal{C}$ into some concrete mathematical object in $\mathbf{Set}$. Now, given a representation F of $\mathcal{C}$ and some $A \in \mathcal{C}$, can I understand the set $F(A)$ using only the categorical information encoded in h_A and F (viewed as abstract functors)? This is a very mathematically useful question to be asking in our situation. And it turns out this question is answered by the Yoneda lemma.

Lemma 1.2 (Yoneda). $\text{Nat}(h_A, F) \cong F(A)$. i.e., the elements of $F(A)$ are in bijection with the natural transformations between h_A and F . Moreover this isomorphism is natural.

This lemma is truly remarkable, since it allows us to describe $F(A)$ using the purely functorial/categorical structure given by h_A and F .

The Correspondence

Now suppose that $\eta: h_A \rightarrow F$ is a natural transformation. Let $X, Y \in \mathcal{C}$ and suppose $f \in \mathcal{C}(X, Y)$. The naturality condition is expressed in the following commuting diagram:

$$\begin{array}{ccc} \mathcal{C}(A, X) & \xrightarrow{f \circ -} & \mathcal{C}(A, Y) \\ \eta_X \downarrow & & \downarrow \eta_Y \\ F(X) & \xrightarrow{F(f)} & F(Y) \end{array}$$

Now suppose that $X = A$ so that the diagram becomes:

$$\begin{array}{ccc} \mathcal{C}(A, A) & \xrightarrow{f \circ -} & \mathcal{C}(A, Y) \\ \eta_A \downarrow & & \downarrow \eta_Y \\ F(A) & \xrightarrow{F(f)} & F(Y) \end{array}$$

Now consider what happens to the identity morphism on A . Going right then down yields $\eta_Y(f)$ while going down then right yields $F(f)(\eta_A(id_A))$. Hence:

$$\eta_Y(f) = F(f)(\eta_A(id_A)).$$

Therefore η is completely determined by $\eta_A(id_A)$. This establishes an injective map $\text{Nat}(h_A, F) \rightarrow F(A)$ by associating each $\eta \in \text{Nat}(h_A, F)$ with $\eta_A(id_A)$. Now we must show that each element of $F(A)$ generates a corresponding natural transformation.

Given an element $a \in F(A)$ and $g \in \mathcal{C}(A, X)$, we define a natural transformation η^a as $\eta_X^a(g) = F(g)(a)$. One must check that this is indeed a natural transformation:

$$\begin{array}{ccc} \mathcal{C}(A, X) & \xrightarrow{f \circ -} & \mathcal{C}(A, Y) \\ F(-)(a) \downarrow & & \downarrow F(-)(a) \\ F(X) & \xrightarrow{F(f)} & F(Y) \end{array}$$

For some $g \in \mathcal{C}(A, X)$, going right then down yields $F(f \circ g)(a)$ while going down then right yields $(F(f) \circ F(g))(a) = F(f \circ g)(a)$. Therefore this is indeed a natural transformation. Furthermore computing $\eta_A^a(id_A)$ yields $F(id_A)(a) = a$. Therefore these processes are inverses and we have established the correspondence between $\text{Nat}(h_A, F)$ and $F(A)$. We will denote this bijection as $\phi_{A,F}(\eta) = \eta_A(id_A)$ and $\phi_{A,F}^{-1}(a) = F(-)(a)$.

Naturality

At this point we have to justify what we mean when we say this correspondence is natural. We can consider the category of set-valued functors on $\mathcal{C}$ whose morphisms are natural transformations. This category is denoted $\mathbf{Set}^{\mathcal{C}}$. Both sides of $\text{Nat}(h_A, F) \cong F(A)$ can be viewed as functors from $\mathcal{C} \times \mathbf{Set}^{\mathcal{C}}$ as:

$$\begin{aligned} (A, F) &\mapsto \text{Nat}(h_A, F), \\ (A, F) &\mapsto F(A). \end{aligned}$$

We then claim that our isomorphism is a natural transformation of these two functors. Spelled out, if we apply a natural transformation in $\mathbf{Set}^{\mathcal{C}}$ to F or a morphism in $\mathcal{C}$ to A , our equation is consistent. We check the first condition: let $\zeta: F \rightarrow G$ be a natural transformation. We must show the following diagram commutes:

$$\begin{array}{ccc} \text{Nat}(h_A, F) & \xrightarrow{\phi_{A,F}} & F(A) \\ \zeta \circ - \downarrow & & \downarrow \zeta_A \\ \text{Nat}(h_A, G) & \xrightarrow{\phi_{A,G}} & G(A) \end{array}$$

Fix some $\eta \in \text{Nat}(h_A, F)$. Going right then down yields $\zeta_A(\phi_{A,F}(\eta)) = \zeta_A(\eta_A(id_A))$. Going down then right yields $\phi_{A,G}(\zeta \circ \eta) = (\zeta \circ \eta)_A(id_A) = \zeta_A(\eta_A(id_A))$ and we are done. Now suppose that $f \in \mathcal{C}(X, Y)$. f generates a natural transformation $h_f: h_Y \rightarrow h_X$ which acts on some $h_Y(W) = \mathcal{C}(Y, W)$ by $- \circ f$. One must check that this is a natural transformation of hom-functors, but I will postpone this to the next section. Now we must show the following diagram commutes:

$$\begin{array}{ccc} \text{Nat}(h_X, F) & \xrightarrow{\phi_{X,F}} & F(X) \\ - \circ h_f \downarrow & & \downarrow F(f) \\ \text{Nat}(h_Y, F) & \xrightarrow{\phi_{Y,F}} & F(Y) \end{array}$$

You know the drill, fix $\eta \in \text{Nat}(h_X, F)$. Going right then down yields $F(f)(\eta_X(id_X))$. Because η is a natural transformation this is just $\eta_Y(f)$. Going down then right yields $\eta_Y(h_f(id_Y)) = \eta_Y(f)$. We are done.

The Yoneda Embedding

Now we want to more deeply understand the hom-functors and how they relate to each other. For this, we introduce the Yoneda embedding. Consider the contravariant functor F from $\mathcal{C}$ to $\mathbf{Set}^{\mathcal{C}}$ given by

$$\begin{aligned} F(X) &= h_X, \\ F(f) &= h_f. \end{aligned}$$

This is the Yoneda embedding. We check that this is a functor. Given composable morphisms $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ in $\mathcal{C}$, we have

$$\begin{aligned} h_f \circ h_g &= h_f \circ (- \circ g) \\ &= - \circ g \circ f \\ &= h_{g \circ f}. \end{aligned}$$

Thus $F(g \circ f) = h_{g \circ f} = h_f \circ h_g = F(f) \circ F(g)$, as required for a contravariant functor.

Furthermore we check that each h_f is indeed a natural transformation. Suppose that $f \in \mathcal{C}(W, Z)$ and $g \in \mathcal{C}(Y, X)$.

$$\begin{array}{ccc} \mathcal{C}(X, W) & \xrightarrow{f \circ -} & \mathcal{C}(X, Z) \\ h_g \downarrow & & \downarrow h_g \\ \mathcal{C}(Y, W) & \xrightarrow{f \circ -} & \mathcal{C}(Y, Z) \end{array}$$

Pick some $k \in \mathcal{C}(X, W)$. Both ways down the diagram yield $f \circ k \circ g$. Therefore h_f is a natural transformation.

Now we claim that the Yoneda embedding is fully faithful.

Definition 4.1. A functor $F: \mathcal{C} \rightarrow \mathcal{D}$ is said to be *fully faithful* if $F_{X,Y}: \mathcal{C}(X, Y) \rightarrow \mathcal{D}(F(X), F(Y))$ is a bijection for each $X, Y \in \mathcal{C}$. If F is a contravariant functor then we instead require $F_{X,Y}: \mathcal{C}(X, Y) \rightarrow \mathcal{D}(F(Y), F(X))$ to be a bijection.

A fully faithful functor $F: \mathcal{C} \rightarrow \mathcal{D}$ says that there is an isomorphic copy of $\mathcal{C}$ contained in $\mathcal{D}$. In our case, F is a contravariant functor, so we are saying that there is an isomorphic copy of $\mathcal{C}^{op}$ contained in $\mathbf{Set}^{\mathcal{C}}$. The proof of this is very simple from the Yoneda lemma. Taking $X, Y \in \mathcal{C}$ and applying the Yoneda lemma to h_Y and h_X gives:

$$\text{Nat}(h_Y, h_X) \cong h_X(Y).$$

However $h_X(Y) = \mathcal{C}(X, Y)$ by definition. This immediately gives us our bijection. Now we are ready to prove the culminating result.

Theorem 4.2 (Isomorphism). Suppose that $h_X \cong h_Y$, then $X \cong Y$.

Suppose that $\phi: h_X \rightarrow h_Y$ is an isomorphism. Because F is fully faithful, it has an inverse F^{-1} defined on $F(\mathcal{C})$. Then $F^{-1}(\phi): Y \rightarrow X$ has inverse $F^{-1}(\phi^{-1})$ and is an isomorphism between X and Y . In the other direction, if $\phi: X \rightarrow Y$ is an isomorphism then $F(\phi): F(Y) \rightarrow F(X)$ is also an isomorphism. It turns out the Yoneda lemma gives us a lot of steam; it is quite powerful.

Rambling

The Yoneda lemma is a very nice lemma and I am glad to have written this note on it. Initially upon studying category theory I was extremely averse to the subject because it felt like useless philosophical ramblings. I finally became invested in category theory when I took Honors Basic Algebra II with Kevin Lin, probably the best math class I've taken. Here, I used category theory and universal properties to prove real mathematical results in an extremely satisfying way. The class really opened my horizons to why mathematicians like category theory. In a sense category theory is the most natural language to talk about mathematics. I myself still have a lot to learn on the topic.